PA194QH6
A new 194-state composition for the consistency of PA

Research prepared for Vladimir Reshetnikov

13 November 2026, Pacific time

Abstract

The requested strict improvement below 194 working states was not obtained. We instead construct the fallback requested in the brief: a binary, single-tape, blank-start machine with 194 working states and a transition graph absent from the supplied corpus. Its name is **PA194QH6**. It combines the premise-exchanged quantifier-distribution emitter with coefficient-six formula constructors and the no-push modus-ponens, cached-formula contradiction-test controller. These component ideas occur in the supplied work; their verified composition is the contribution here.

The 202-control parent has 135 fixed controls and 67 axiom readers. Eight simultaneous contextual folds leave 194 controls. The complete 3,775-site source graph passes 4,876 universal outcome contracts in each of two implementations. The second checker rederives its register-list lemmas from literal transitions. Blank execution reaches its exact clean source frame after 58,088 steps. The comparison uses a pinned repository snapshot, the 51 placed research packets, and all four current incoming archives, with a separate accounting of rejected and auxiliary machines.

An ordinary mathematical argument gives PA $\vdash \neg\,\mathrm{Halt}(M_{\mathrm{QH6}}) \leftrightarrow \mathrm{Con}(\mathrm{PA})$. It explicitly composes the supplied syntactic PA calibration, a finite proof translation for exchanging the quantifier axiom's premises, the command compiler, and the physical simulation. The inherited calibration is identified rather than silently replaced by standard-model soundness. No newly completed end-to-end Lean or Rocq proof, no iPRA endpoint, and no 194-state lower bound are claimed.

Contents

1 Result, machine convention, and scope **3**

2 Comparison of the supplied designs **3**

2.1 Frozen scope and the meaning of a distinct machine 3

2.2 The principal construction families 4

2.3 Why the delivered table is materially different 5

3 Literal construction and the eight folds **6**

4 The numerical calculus **7**

5 Calibration to PA and its inherited dependencies **8**

5.1 Formula codes and variable labels must move together 8

5.2 The numerical dominance test is not preserved by recoding 9

5.3 Soundness is a syntactic proof transformation 9

6 The command compiler and exhaustive search **10**

6.1 The actual store interface 10

6.2 Explicit compiler words 10

6.3 The scheduler carries its store 11

7 Physical verification and what its numbers mean **12**

8 Attempts below 194 and the remaining obstacle **12**

9 What transfers from ZFC and ConciseIPRA **13**

10 Reproduction and claim ledger **14**

Sources and attribution **15**

A Complete dense transition table **16**

B The 194-state frontier **18**

C All 55 research deliveries **20**

1 Result, machine convention, and scope

The deliverable contains the literal table `machine/machine.tm` and its native-label companion. The dense presentation has working states $0, \dots, 193$, initial state 0, and a separate terminal label 194. There are exactly 388 rows. A row

$$q \quad b \quad w \quad D \quad q'$$

means: in state q, reading b, write w, move one cell in direction $D \in \{L, R\}$, and enter q'. There is no row for the terminal label. The head starts at cell zero on the all-zero, two-way infinite tape. Initialization is included in the state count.

All 194 working controls are reachable in the pointed transition graph. That finite graph fact does not assert that every control is visited by the blank computation. The native presentation starts at 74 and halts at 254; the supplied label list is checked as an exact transition conjugacy, including the designated start and halt.

The dense table's SHA-256 is

`a48c4729cb35c298c364ba5a7c1d5e489ea3c111f482a93e57338d4c9847d1e9`.

Let H_M be the arithmetic sentence asserting the existence of a finite halting computation from the specified blank start, with a standard primitive-recursive finite-trace verifier. The claimed calibration is the following ordinary mathematical result.

Theorem 1.1 (PA calibration)**.** *With the literal table just specified,*

$$\mathrm{PA} \vdash H_{M_{\mathrm{QH6}}} \leftrightarrow \exists p \, \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner), \qquad \mathrm{PA} \vdash \neg H_{M_{\mathrm{QH6}}} \leftrightarrow \mathrm{Con}(\mathrm{PA}).$$

The proof uses the explicitly retained numerical-calculus calibration described in Sections 4–5.

The theorem does not assume Con(PA). That assumption is needed only for the further external conclusion that the particular blank machine never halts. The executable certificates concern finite syntactic and symbolic objects; they are not themselves a formal proof of the displayed arithmetical theorem. The precise boundary is recorded in Section 10.

2 Comparison of the supplied designs

2.1 Frozen scope and the meaning of a distinct machine

The comparison is pinned to ProveIt commit

`2e00058366845a6e2e0368004fd9ed2050a06a9a`.

The primary paths are `Computability/BusyBeaver/Research/PA` and `docs/incoming`. The former already contains 51 filed research packets. The latter has exactly four further archives at this snapshot: `PA194NS_research.zip`, `pa194h6_conpa_2026-11-13.zip`, `pa194haz_research_2026-11-14.zip`, and `pa194qn_research_2026-11-13.zip`. Thus the delivery ledger has 55 entries. The ZFC and ConciseIPRA trees are inspected as possible sources of construction ideas, not counted as PA-machine deliveries.

The repository's historical census concerns 4,984 original table files. Most have since been removed as duplicates. Its 490 classes include 444 classes with a resolved start and 46 table classes without

one. Those are historical accounting numbers, not the number of machine files currently present. Our current scan reads the retained files and the current archives, recursively including nested reference archives, and records each recognized complete numeric binary transition table. The audit artifacts retain the scope and exact hash of every input.

Graph comparison preserves read symbols, write symbols, moves, and the halting control. State numbers and row order have no significance. The candidate is additionally checked against global tape reflection and global exchange of the binary symbols. A graph difference is only a necessary part of the novelty argument: an edit to an unused transition would not by itself establish a materially different executed design.

The current census finds 1,445 recognized table occurrences, 725 byte forms, and 499 complete one-halt action-graph classes with a spanning root. The 499 equals the repository's 490, less three small diagnostic fixture classes, plus twelve new classes in the current incoming archives. Of the current classes, 40 have 194 working states: 27 have supplied ordinary PA-equivalence manuscripts, nine are auxiliary/search tables, and four are explicitly rejected. No eligible full consistency-candidate table in the scanned literal formats has fewer than 194 states.

The location counts are 506 direct PA files, 127 table occurrences in nested PA ZIPs, and 812 in the current incoming archives. Their byte forms and graph classes overlap heavily. The scanner recognizes numeric five-column tables in `.tm`, `.txt`, `.tsv`, and `.csv`; it does not count unmaterialized parameter families or JSON-only machine encodings as literal tables. Named contributions outside that encoding are still represented in the delivery ledger. The small diagnostic fixtures are recorded separately, not treated as consistency-machine candidates.

PA194QH6 matches none of the 40 pre-existing 194-state classes. The novelty checker compares full canonical tuples, not only their hashes, under arbitrary state renaming, every spanning-root choice, reflection, and binary-symbol exchange. This requires 160 exact comparisons after considering the four candidate symmetries. The supplied primary PA194 is also used as a positive control: the checker correctly finds its existing class. See `comparison/QH6_novelty.json` and `comparison/novelty_positive_control.json`.

The annotated 194-state frontier and the complete 55-delivery ledger are included in the appendices and in machine-readable files. Rejected scanner contractions and diagnostic tables are kept separate from proposed consistency machines. Counts of tables or graph classes must not be read as counts of proved consistency equivalences.

2.2 The principal construction families

Family	States	What changes, and what must be proved
U208 / U209	208–209	Earlier source/compiler route. Several deliveries repeat the same graphs and the same contextual quotient; they do not supply distinct lower-state improvements.
Imported PA196	196	The main register-engine and numerical-calculus baseline. Its formal result includes the PA implication from nonhalting to consistency. The opposite internal direction is not supplied merely by its metatheoretic nonhalting theorem.

Family	States	What changes, and what must be proved
PA195 families	195	Reassociated axiom readers, separated diagonal head, alternate register placement, cached-result controller, and scanner experiments. Some apparent 194-state descendants in these packets are explicitly rejected.
Primary PA194	194	Preserves the 190-token separate-head emitter and its full source graph; 134 fixed states plus 67 reader states minus seven contextual folds.
C / CS / R	194	Change where an MP result is stored, the Shift controller, or the arithmetic/comparison arrangement. These require a command compiler for the changed store interface.
N / H / HS / NS	194	Omit the successful-MP stack push, or test only the cached formula for contradiction. The successful MP leaves its conclusion in the offset register; overlooking that fact invalidates a proposed proof compiler.
Radix 5 / 6	194	Replace $4p$ by $5p$ or $6p$ for formulas, retaining ordinary Cantor pairing for stacks and enumeration. The proof must recode variable labels as well as formulas and handle unused residues.
Q / QN	194	Exchange the curried premises of the universal-distribution head. Q uses the cached-result controller; QN uses no-push MP with the sum-based Shift test.
D	194	Emits the separate diagonal head at $\Gamma(D)$ instead of $\Gamma(A)$, requiring a different selector; its layout uses 69 readers and nine folds.
J / ZR / HAZ	194	Alter source-tree control or replace a jump by a decrement at a proved-zero register. A local invariant and the actual resolved successor map are essential.
QH6, here	**194**	**Exchanged distribution, coefficient six, no-push MP, cached-only Shift.** The contribution is a newly verified composition, with a new literal graph and a complete source/table binding.

2.3 Why the delivered table is materially different

The closest designs differ along actual source operations:

Machine	Formula factor	Distribution	Successful MP	Shift halt test
Primary PA194	4	original	push and return result	$t + h = 1$
PA194Q	4	exchanged	return result in cache	$h = 1$
PA194QN	4	exchanged	no push; top result	$t + h = 1$
PA194H6	6	original	no push; top result	$h = 1$
PA194QH6	**6**	**exchanged**	**no push; top result**	$\boldsymbol{h = 1}$

The coefficient change affects every formula constructor call and MP's numerical implication comparison. Exchanging distribution changes a formula actually emitted by AX. The no-push/cached-test interface changes the recurrent proof-search store protocol. The resulting source is not obtained by renaming control states or mirroring the tape. Nevertheless, each component idea is credited to the earlier packets; combining them does not constitute a new strict state reduction.

For a concrete numerical witness, set $A = 1, D = H = 0$. Then $P = U(0, 1) = 8$, $Q = U(0, I(1, 0)) = 128$, and $R = U(0, 0) = 2$. The original H6 distribution head $I(P, I(Q, R))$ has code 7,837,156,242, whereas QH6 emits $I(Q, I(P, R)) = 666,162$. This is an actual AX output difference, not an edited unused transition.

A naming collision matters in this corpus. The filed PA194NS coincides with the filed PA194HS, whereas the latest `PA194NS_research.zip` uses PA194NS for a different sum-test machine. The comparison ledger disambiguates these by packet and hash, rather than assuming a name is a machine identity.

3 Literal construction and the eight folds

The builder begins with the frozen Q emitter grammar and the existing register engine. The coefficient-six constructor is assembled from a six-increment subtree in place of the four-increment subtree. MP's success path omits the zero push; the Shift path clears the old top and tests the cached value. All changes are represented both as source-tree definitions and as literal transition rows. The source resolver checks the opcode, register operand, and every successor, including decrement's two branches.

Against PA194H6, exactly 83 of the 3,775 resolved instructions change and 3,692 remain identical. The emitted macro word changes only in the 19-token distribution interval. A separate relocation certificate compares the four-increment and six-increment sources: 61 multiplier bodies widen by two increments, all 3,409 sites outside the old bodies keep their operations and relocated successors, and no external edge enters a body's interior. The loop invariant for a six-increment body with initial counter m is $X = m - j$, $T = T_0 + 6j$. This proves the coefficient change for arbitrary counter values. The exact bindings and the inherited generic constructor proof are expanded in `proof/qh6_preservation.tex` and the retained references.

The parent has 135 fixed controls and 67 AX reader controls. Its size is therefore 202, and eight simultaneous quotients give

$$\boxed{135 + 67 - 8 = 194.}$$

The eighth fold joins fixed state 41 to state 99. It is valid in the actual QH6 table, together with the seven reader folds; it is not inferred merely from a successful check on a different parent.

Removed	Retained	Action on 0	Action on 1
2004	2052	$0L2015$	$1L2048$
2021	2030	$0R90$	$1L2023$
2025	2035	$0L2002$	$1R61$
2027	2046	$0L2036$	$1R2031$
2039	2044	$0R2040$	$1L2014$
2055	2056	$0L2042$	$1R2043$
2058	2017	$0L2050$	$1L2062$
41	99	$0L30$	$1L2040$

For a fold, the two old controls have different known incoming history bits. The retained control may move back to read that bit, then dispatch through an already existing helper and resume the old computation. The certificate exhausts each history/current-symbol combination. It checks the entire touched tape fragment, the head displacement, and the mapped exit state in the final simultaneously folded table. The exterior tape is arbitrary.

Lemma 3.1 (Contextual quotient)**.** *From the specified source frames and blank initialization, finite parent runs and finite quotient runs simulate one another at the chosen boundary configurations. In particular they have equivalent halting statements, provably in PA.*

Proof. Outside folded controls the rows agree under the quotient map. At a folded control its incoming transition establishes the history-bit condition. The finite local certificate supplies the matching positive-length path, preserves the exterior tape, and reaches the prescribed next boundary. The certificate is checked after all folds, so helper paths are not assumed to retain behavior that another fold has changed. Induction on simulated boundaries gives the forward translation. The checked boundary/exit contracts and the positivity of every segment exclude an extra quotient halt or an infinite internal segment. These are finite coding arguments and induction on natural trace lengths, hence formalizable in PA. □

The exact blank simulation reaches the clean source entry at native state 84 with head position -2, after 58,088 transitions. The logical register vector is $(2, 0, 0, 0, 0, 0, 0, 0, 0)$, with the stated permutation into physical blocks and 26 unit spare blocks. The checker compares the complete finite support of the tape, not just a displayed window.

4 The numerical calculus

Put

$$p(a, b) = \frac{(a + b)(a + b + 1)}{2} + b,$$

and define

$$I(A, B) = 6p(A, B), \quad M(a, b) = 6p(a, b) + 1, \quad U(a, B) = 6p(B, a) + 2, \quad E(a, b) = 6p(a, b) + 3.$$

Code zero denotes truth and code one denotes the contradiction $M(0, 0)$. Write $N(B) = I(B, 1)$, $A \wedge B = N(I(A, N(B)))$, and $\exists x_a B = N(U(a, N(B)))$. Implication in the following displays means the numerical constructor I. Its exceptional equality $I(0, 0) = 0$ is accounted for; the representation is not a free syntax algebra at zero.

The calculus $\mathcal{C}_6^{\mathrm{Q}}$ has modus ponens and generalization of theorems. There are no undischarged hypotheses in its generalization rule. Its six universally instantiated heads are

$$\begin{aligned}
h_1(D, H) &= (\exists x_D U(D, H)) \to H, \\
h_2(D, A, H) &= U(D, U(A, H)) \to U(A, U(D, H)), \\
h_3(D, A, H) &= M(H, D) \to (E(H, A) \to M(A, D)), \\
h_4(D, A, H) &= E(H, D) \to (M(A, H) \to M(A, D)), \\
h_5(D, H) &= \exists x_D E(H, D), \\
h_6^{\mathrm{Q}}(D, A, H) &= U(D, A \to H) \to (U(D, A) \to U(D, H)).
\end{aligned}$$

The last head is the premise-exchanged one. The original last head is $U(D, A) \to (U(D, A \to H) \to U(D, H))$.

The dominated propositional head is

$$\rho(D, A, H, u) = ((H \to u) \to D) \to ((D \to H) \to (A \to H)), \qquad u \le D.$$

There is a separate diagonal head $\Gamma(B) = B \to U(B, B)$. Finally, put

$$\begin{aligned}
V(a,b,c) &= M(a,b) \to U(c, M(a,b)),\\
P(a,b,c) &= c \to M(c,b),\\
S(a,b,c) &= U\big(b, (M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big),\\
Q(a,b,c) &= U(c, S(a,b,c) \to N(P(a,b,c))),\\
G(a,b,c) &= N(V(a,b,c) \to Q(a,b,c)).
\end{aligned}$$

The guarded gate is $G(a, b, c)$ when $c \neq a, b$; a failed guard returns the already derivable zero. The letter Q in this gate formula is inherited notation and is unrelated to the suffix in PA194QH6.

Lemma 4.1 (Exchange of distribution)**.** *The coefficient-six original and exchanged-head calculi have primitive-recursive proof translations preserving every conclusion. PA proves the correctness of both translations.*

Proof. In the inherited implicational calculus there is a fixed finite derivation of

$$(X \to (Y \to Z)) \to (Y \to (X \to Z)).$$

One can obtain it using only the derived K and S axiom schemas and modus ponens: abstract the elementary deduction that applies a hypothesis $X \to (Y \to Z)$ first to X, then to Y, in the order of assumptions Y, X. The included exchange certificate checks a literal finite Hilbert derivation. Applying the displayed template to each old distribution head yields the new one. Applying it with X, Y exchanged yields the old head from the new one. All other axiom heads and both rules are unchanged. Replace axiom leaves by these templates and induct on proof trees. There is no appeal to a semantic completeness theorem or to a new substitution rule for freshness-restricted axioms. Here K and S are derived solely from the unchanged restricted ρ basis, using reflexivity and the bounds $a, b \leq I(a, b)$; the altered quantifier-distribution head is not used to justify its own replacement. □

5 Calibration to PA and its inherited dependencies

The supplied separate-head coefficient-four calculus has explicit mathematical constructions translating PA refutations into numerical refutations and numerical refutations back into PA refutations. The retained proof sources document the equality-rewriting and neighborhood interfaces, the diagonal replacement, and the finite binary-membership interpretation. Those are substantial inherited lemmas, not results of the new transition search. The following explains the extra issues needed to use them here.

5.1 Formula codes and variable labels must move together

For general coefficient $k \geq 4$, define $F_k(0) = 0$ and, for $n > 0$,

$$n = 4p(a,b) + r \ (0 \leq r < 4) \quad \Longrightarrow \quad F_k(n) = kp(F_k(a), F_k(b)) + r.$$

The recursive arguments are smaller than n. Course-of-values recursion can be implemented by ordinary primitive recursion storing the previously computed sequence. PA proves the defining identities and positivity. A similar decoder with the coefficient changed back to four, mapping unused residues to zero, is a left inverse on the image. Thus F_k is injective, fixes 0 and 1, respects all four constructors, and preserves the equality/inequality guards among labels.

Recoding a variable label a as $F_k(a)$, as well as recoding formulas, is essential. In particular,

$$F_k(\Gamma_4(B)) = \Gamma_k(F_k(B)).$$

Holding labels fixed would lose this identity, since the diagonal binder is the formula's code itself. Binding and shadowing are preserved by the same injective renaming.

5.2 The numerical dominance test is not preserved by recoding

The recoding is not monotone. For example,

$$40 < 44, \qquad F_5(40) = 75 > 55 = F_5(44).$$

Accordingly an old dominated ρ axiom cannot simply be recoded and declared a new dominated axiom. The retained `RestrictedImplication.lean` supplies the finite theorem `rho8All`: the unrestricted implicational schema follows from the dominated schema whenever the dominance relation is reflexive and extends from either argument to an implication containing that argument. Here the required inequalities follow from

$$a, b \le p(a, b) \le kp(a, b) = I_k(a, b).$$

Each old dominated leaf is replaced by this fixed unrestricted derivation at the recoded parameters. Its own restricted leaves satisfy the numerical inequalities. This resolves the order obstruction without assuming it away.

5.3 Soundness is a syntactic proof transformation

Decode a positive coefficient-six code $6p(a, b) + r$. Translate tag 0 by implication, tag 1 by arithmetic binary membership $\mathrm{Bit}(x_a, x_b)$, tag 2 by $\forall x_b\, T(a)$, and tag 3 by $x_a = x_b$. Translate zero and tags 4 and 5 by closed truth. The unused residues must be included because the scheduler supplies arbitrary numerical parameters.

Every source label in $T(B)$ is strictly below B; auxiliary bound variables for the arithmetic bit definition are chosen capture-free. Thus x_B is not free in $T(B)$, proving the diagonal head by a uniform first-order proof template. The alias $I(0, 0) = 0$ uses a fixed proof of truth equivalent to truth implying truth.

The pointed gate is supplied by finite binary comprehension. For every arithmetic formula $\theta(z, \vec{u})$, PA has the induction proof

$$\forall n\, \exists s < 2^n\, \forall z < n\, (\mathrm{Bit}(s, z) \leftrightarrow \theta(z, \vec{u})).$$

At the successor step retain s or add 2^n, according as $\theta(n, \vec{u})$ fails or holds. Substituting the syntax of θ into this fixed induction pattern is primitive recursive. This is a construction of PA proofs, not a PA truth predicate or a reflection axiom. The inherited gate proof treats the collision $a = b$ separately, because the binder then changes both occurrences of that variable label.

Combining this interpretation with hereditary recoding and Lemma 4.1 yields primitive-recursive proof constructors α, β whose correctness is provable in PA:

$$\mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner) \Longrightarrow \mathrm{Prf}_{\mathcal{C}_6^{\mathbb{Q}}}(\alpha(p), 1),$$
$$\mathrm{Prf}_{\mathcal{C}_6^{\mathbb{Q}}}(q, 1) \Longrightarrow \mathrm{Prf}_{\mathrm{PA}}(\beta(q), \ulcorner 0 = 1 \urcorner).$$

These are the logical interfaces consumed by the machine proof.

6 The command compiler and exhaustive search

6.1 The actual store interface

A command store is (t, w, d, h): current formula, Cantor stack, numerical offset, and cached formula. Write $[a; w] = p(a, w)$. Both total projections of zero are zero, and $[0; 0] = 0$; the stack is not a free list.

For $b = d + x$, QH6 has the exact interfaces

$$\mathsf{MP}_x(t, w, d, h) = \begin{cases} (b, w, b, 0), & I(t, b) = h, \\ (0, w, b, 0), & I(t, b) \neq h, \end{cases}$$

and

$$\mathsf{S}(t, w, d, h) = \begin{cases} \mathsf{halt}, & h = 1, \\ (h, \mathrm{tl}(w), d, \mathrm{hd}(w)), & h \neq 1. \end{cases}$$

Successful MP leaves $d = b$, does not insert a dummy zero into the stack, and clears the cache. These details determine the proof compiler.

For $D = d + x, H = h + y, A = c$, the AX command has output

$$\mathsf{AX}_{c,x,y}(t, w, d, h) = (G(H, D, A), [q_1, \ldots, q_9, t; w], 0, 0),$$

with the guard understood, where

$$(q_1, \ldots, q_9) = (\Gamma(A), h_6^{\mathrm{Q}}, h_5, h_4, h_3, h_2, h_1, \rho(D, A, H, d), U(d, h)).$$

The head parameters are those displayed in Section 4. The complete constructor trees and their order are checked symbolically. The first eight heads and the gate are zero modulo six; the ninth is two modulo six and at least two. None is code one. The saved old top t is not included in that assertion.

6.2 Explicit compiler words

Let $A_0 = \mathsf{AX}_{0,0,0}$, with words executed left to right. Define

$$\mathsf{Push} = A_0; \mathsf{S}^9; \mathsf{MP}_0, \quad \mathsf{Pop} = A_0; \mathsf{S}^{11}, \quad \mathsf{Discard} = \mathsf{MP}_0, \quad \mathsf{Offset}_i = \mathsf{MP}_i.$$

For arbitrary displayed stores the useful identities are

$$\begin{aligned} \mathsf{Push}(t, w, d, h) &= (0, [t; w], 0, 0), \\ \mathsf{Pop}(t, [a; w], d, h) &= \begin{cases} \mathsf{halt}, & t = 1, \\ (t, w, 0, a), & t \neq 1, \end{cases} \\ \mathsf{Discard}(t, w, 0, h) &= (0, w, 0, 0), \\ \mathsf{Offset}_i(0, w, 0, 0) &= (0, w, i, 0). \end{aligned}$$

Push's last comparison fails because implication and universal-head codes have distinct residues. In Discard, the two MP outcomes agree when the candidate conclusion is zero. Pop's eleventh Shift tests the saved old top. These arguments include the canonical zero-stack cases.

An axiom head q_j, for $1 \leq j \leq 7$, is selected from a clean store by

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; \mathsf{S}^{j+1}; \mathsf{Pop}^{9-j}.$$

The guarded gate is selected by $\mathsf{Discard};\mathsf{AX}_{c,b,a};\mathsf{Pop}^{10}$. For $u \le a$, the dominated head uses

$$\mathsf{Discard};\mathsf{Offset}_u;\mathsf{AX}_{b,a-u,c};\mathsf{S}^9;\mathsf{Pop}.$$

Generalization at index i uses

$$\mathsf{Gen}_i = \mathsf{Push};\mathsf{Offset}_i;\mathsf{S};\mathsf{AX}_{i,0,0};\mathsf{S}^{10},$$

which returns $(U(i,a),w,0,0)$ from $(a,w,0,0)$.

At an MP proof node with subproofs of a and $I(a,b)$, use

$$\begin{aligned} W_p = {} & \mathsf{Discard};\mathsf{Push};W_{\text{major}};\mathsf{Push};W_{\text{minor}};\mathsf{Push};\\ & \mathsf{S};\mathsf{S};\mathsf{MP}_b;\mathsf{Pop}. \end{aligned}$$

Immediately before the last two Shifts the store is $(0,[a,I(a,b),0;w],0,0)$. The Shifts expose the two premises. Successful MP yields $(b,[0;w],b,0)$; Pop consumes the reserved zero and restores the offset and cache, or halts upon detecting $b=1$. The compiler, not MP, supplied that zero.

Induction on the proof tree shows that W_p, from every $(t,w,0,0)$, either halts or returns $(\mathrm{concl}(p),w,0,0)$. If the conclusion is one, the word

$$A_0;W_p;\mathsf{Pop}$$

halts from *every* incoming store. The initial AX establishes the clean-offset/cache precondition. A plain final Shift would be wrong: $\mathsf{S}(1,0,0,0)=(0,0,0,0)$. The final Pop is essential.

6.3 The scheduler carries its store

The unchanged scheduler enumerates finite command words without resetting the theorem store. The all-store compiler just proved is therefore needed. One explicit continuation convention is

$$E(\epsilon)=0, \qquad E(c::w)=1+B_c(E(w)),$$

where

$$B_{\mathsf{S}}(z)=p(0,z), \quad B_{\mathsf{MP}_x}(z)=p(1,p(x,z)), \quad B_{\mathsf{AX}_{c,x,y}}(z)=p(c+2,p(y,p(x,z))).$$

If z codes the all-store halting word, prefix it by $\mathsf{AX}_{k,0,0}$. The resulting initial candidate values

$$n_k = p(k+2,p(0,p(0,z)))$$

are increasing and unbounded. The prefix can change the store arbitrarily; the remainder still halts. Positive continuations decrease while decoding. The arithmetic command loops have primitive-recursive termination bounds. PA consequently proves that one of these witnesses is reached, unless an earlier trial already halts.

For the converse, annotate top, cache, and each live canonical stack entry with a proof in $\mathcal{C}_6^{\mathrm{Q}}$. Zero has a fixed proof and supplies the virtual zero tail. AX, MP, and Shift preserve the annotations; the only halt exposes an annotated cache equal to one. Primitive recursion on a finite run therefore extracts a numerical refutation. Compose with α,β and the physical simulation to prove Theorem 1.1.

7 Physical verification and what its numbers mean

Checked quantity	Observed result
Working states / transition rows	194 / 388
Parent fixed controls / AX readers / folds	135 / 67 / 8
Resolved source sites	3,775
Primitive source outcome contracts	4,876
Construction checker symbolic steps	1,601,532
Construction checker maximum case length	499
Independent literal checker symbolic steps	1,169,239
Independent checker maximum case length	403
Independent minimum positive-duration constant	3
Literal blank initialization steps	58,088

The 4,876 outcomes include both branches of each decrement. The variables range over all natural register values, not a selected finite range. The symbolic frame includes an affine header gap, the full register file, and an arbitrary nonempty finite spare list. The independent checker derives four register-list lemmas directly from literal transition rows, resolves source control with a second implementation, and checks native/dense conjugacy. It does not call the construction checker's arithmetic-box oracle.

More precisely, for source address p, the frame gap has the form $48 + n - |p|$, with $n \geq 0$ and the verified bound $|p| \leq 36$. This is the stated domain of the affine symbolic proof, rather than an assertion about arbitrary negative gap lengths or arbitrary malformed tapes.

The displayed symbolic step counts are counts of checker operations, not uniform bounds on TM runtime. An individual symbolic list scan represents arbitrarily many literal steps. Its positive-duration formula and list invariant justify that abstraction. Both the local fold checks and the whole-source contracts are needed: a plausible fold can break a distant source path, and a bounded blank-run prefix cannot validate arbitrary register values.

Executable symbolic certificates still have a trust boundary: their Python implementations, the mathematical soundness of their symbolic rules, and the explicitly reused source-calculus calibration. They provide substantially more evidence than random tests, but are not a single theorem checked by a proof-assistant kernel. The report keeps that distinction throughout.

8 Attempts below 194 and the remaining obstacle

A fresh whole-instruction quotient screen considered 62,073 row-choice merges on PA194QH6. Every candidate failed at least one of the 4,876 source-outcome contracts under the checker's 1,024-step symbolic budget. The unmodified machine passes the same contract set. This is rejection by the stated checker and fuel bound, not a theorem that every failed proposal is semantically inequivalent.

A broader local search allowed arbitrary replacement first actions, with radius-five tape windows and at most 150 local transitions. An initial source/startup context collection contained 16,937 windows. A stronger collection derived from the independent literal executor, all four generic register-list inductions, and complete startup contained 30,761 windows covering all 194 controls. No compatible merge or single-state elimination was found in either collection. The stronger run also allowed removal of the initial control 74. This control is used again inside the increment engine; its initial role does not make it disposable.

A source-preserving rotation of the main binary control tree lowers the leftmost source address from six zero bits to five. Resolving the entire 3,673-site primary source graph confirms exact control-flow identity for that rotation. However, the proposed shorter scanner still fails 52 source contracts: a changed predecessor bit breaks a contextual dispatch, and long zero runs remain in the AX subtree. Thus neither source graph identity alone nor shortening one address certifies the scanner deletion.

A fresh fixed-word reassociation search used the exact QH6 fixed table and recomputed source attributes for the six-increment multiplier. It allowed compatible fixed/reader folds and ran for 120.003 seconds with seed 202611140617. It considered 1,386,441 mutation slots and accepted 52,092 proposals. Its best effective reader-minus-fold score remained 59, giving $135 + 59 = 194$ total controls, with no sub-194 nomination. The source, seed grammar, attributes, final search population, and actual log are in `research/grammar/`. Wall-time-limited search counts are observations of this run, not guaranteed identical across different machines.

The accompanying reduction receipts and portable replay scripts are in `research/reduction/`. The spare-append-policy search was interrupted and is excluded from the completed counts. None of these unsuccessful searches is used as an assumption in the accepted machine's proof.

The construction must be optimized as a whole. Counting the best reader grammar and the best folds from different candidates independently can give a fictitious state saving. Likewise, reducing an expanded instruction count need not remove a physical control: shared macros can retain that control for another purpose.

A stronger freshness guard $c > \max(a, b)$ initially looks simpler than two inequalities $c \neq a, b$. It does not immediately inherit the existing logical proof. The reverse-rewriting argument uses membership vacuity at a label smaller than the self-fresh separator code, so imposing a numerical ordering on every such instance creates conflicting requirements. Padding a formula by a tautology can make a new separator code larger, but that alone does not recover every vacuity interface that the proof compiler uses. No unsupported guard-deletion candidate is delivered.

These results are bounded negative evidence. They neither prove optimality of 194 states nor exhaust alternative arithmetic encodings, source calculi, or nonlocal tape invariants. The fallback table is provided because the strict target was not achieved, as the request permits.

9 What transfers from ZFC and ConciseIPRA

The useful ZFC ideas are control-tree sharing, context-sensitive quotients, explicit primitive-recursive proof translation, and separate validation of the logical source and its physical compiler. These methods already underlie the strongest PA descendants, and they are retained here. Merely choosing a small ZFC consistency machine does not solve the PA task: its nonhalting expresses a different consistency statement, and deleting a source axiom requires a new proof of the resulting arithmetic strength.

ConciseIPRA demonstrates a productive design principle: remove an expensive primitive interface when a fixed derived construction can replace it. It also stresses whole proof-checker complexity, rather than only the number of displayed logical symbols. The exchanged quantifier axiom here fits that discipline: its equivalence is witnessed by a finite proof template, while the actual gain or cost is measured in the compiled TM. Replacing PA by iPRA simply because an iPRA checker is short would change the target; no PA–iPRA equiconsistency is assumed.

In this investigation these trees contributed methods and constraints. No new ZFC or ConciseIPRA theorem is claimed, and no uncompiled alternative is advertised as a smaller machine.

10 Reproduction and claim ledger

The archive is self-contained for the delivered checks. It contains the literal machine, deterministic builder, source grammar and resolved source, fold witnesses, both physical checkers, command and exchange certificates, the pinned comparison census, search receipts, this article's sources and PDF, and the relevant inherited proof sources and licenses.

From the archive root, with Python 3.10 or later and its standard library:

```
python3 -B verify.py --receipt fresh-verification.json
```

The default rebuilds the machine in a temporary directory, checks byte identity with the delivered artifacts, verifies command words and exact source relocation, rechecks the 61-line implicational exchange certificate, compares the candidate against the frozen graph census, and runs the independent literal-tape checker. No network, source checkout, external solver, or proof assistant is required. Keep assertions enabled; do not use Python's optimization flags.

`--quick` omits only the independent physical replay, and reports that omission explicitly. The full replay receipt delivered in the archive was produced with the default, including that replay. The inputs are hash-pinned; the supplied read-only checks write temporary or explicitly requested outputs.

To rebuild just the machine, run

```
python3 -B construction/build_qh6.py --output rebuilt-machine
```

To regenerate the full current-corpus census, rather than check novelty against its frozen output, the separate audit script requires a checkout of the pinned repository:

```
python3 comparison/audit_corpus.py --repo /path/to/ProveIt \
  --out fresh-corpus-audit
```

It rejects a different HEAD and verifies materialized source bytes against the pinned Git blobs. The census input corpus itself is not duplicated in this compact archive; exact canonical transition tuples are included for offline novelty checking.

The PDF can be regenerated with a TeX Live installation providing the packages listed in `report/main.tex`:

```
cd report
pdflatex -interaction=nonstopmode -halt-on-error main.tex
pdflatex -interaction=nonstopmode -halt-on-error main.tex
```

The optional bounded-search programs and receipts are under `research/`. Their failure to find a smaller table is independent of the deterministic acceptance checks for the delivered machine.

Claim	Status
194 states, binary blank start, 388 rows	Directly checked against the delivered literal table.
Graph absent from the supplied 194-state corpus	Exact finite graph comparison, with state renaming and extra symmetry checks; the declared file-format scope is retained.

Claim	Status
Material source difference	Explicit formula-code, emitted-head, and recurrent command-interface differences; not inferred solely from a hash.
Universal source-to-tape contracts	Passed in two symbolic implementations, including a literal-row register-list audit and complete blank startup.
PA-provable consistency equivalence	Ordinary mathematical composition of explicitly identified inherited PA calibration, exchanged-head proof translation, command compiler, scheduler argument, and physical contracts.
New end-to-end Lean/Rocq theorem	Not supplied. Existing repository theorems do not by themselves discharge every machine-specific calibration obligation.
An iPRA proof of the equivalence	Not claimed; it is a separate, stronger formalization goal in the current repository.
A machine below 194 states	Not obtained. Rejected contractions are research artifacts, not alternative solutions.
A lower bound of 194 states	Not established.

Sources and attribution

All repository references below use the frozen snapshot named in Section 2. The packet ledger records exact relative paths and source hashes. The central dependencies are:

1. ProveIt, `Computability/BusyBeaver/Research/PA/README.md`, for the current formalization status and the links among the source, literal machine, and arithmetic endpoints. Pinned PA tree.
2. James Chang's imported PA196 development, especially `PA196/Program.lean`, `M6/Runtime.lean`, `M6/Logic/RestrictedImplication.lean`, and its formal-result summary.
3. The primary `pa194_conpa_2026-11-13` packet, for the separated-head calculus, PA calibration, reader construction, and independent physical-checking method.
4. The `pa194q_conpa_2026-11-13` packet, for the exchanged quantifier-distribution emitter and its finite implicational proof.
5. The `pa194_radix_family_2026-11-13` and current `pa194h6_conpa_2026-11-13` packets, for hereditary recoding, coefficient-six arithmetic, the hybrid controller, and its all-store proof compiler.
6. The remaining placed and incoming packets, individually identified in the ledger, for the comparison frontier, earlier alternatives, and rejected experiments. Pinned incoming archives.
7. ProveIt, `Computability/BusyBeaver/Research/ZFC` and `Logic/PrimitiveRecursiveArithmetic/ConciseIPRA`, for the additional architectural ideas assessed in this report.

The retained source licenses and provenance distinguish imported work from the new composition and checks. This report does not claim that existing component constructions were invented here.

A Complete dense transition table

Here H means terminal label 194. An action such as $1R27$ writes 1, moves right, and enters state 27. Start in state 0 on the all-zero tape.

State	Read 0	Read 1	State	Read 0	Read 1
0	$1R1$	$1R2$	97	$0R36$	$1R114$
1	$1R3$	$1R1$	98	$0R113$	$1R75$
2	$0R0$	$1R2$	99	$0L63$	$0R115$
3	$0R4$	$0R1$	100	$0L62$	$1R116$
4	$1R5$	$1L6$	101	$0L90$	$1R117$
5	$1L7$	$1L8$	102	$0R118$	$0L14$
6	$0L7$	$1L6$	103	$0R69$	$1R103$
7	$0L9$	$1L6$	104	$0R119$	$1R95$
8	$0L10$	$0L11$	105	$0R119$	$1R67$
9	$1R12$	$0L9$	106	$0R120$	$0L14$
10	$0L9$	$0L10$	107	$0R121$	$1L71$
11	$1L8$	$1L11$	108	$0R122$	$1R123$
12	$0R12$	$0R13$	109	$0R112$	$1R109$
13	$0L14$	$0R13$	110	$0R124$	$1R125$
14	$1L15$	$0L15$	111	$0R109$	$1R111$
15	$0L15$	$1L16$	112	$0R115$	$1R112$
16	$0L17$	$1L16$	113	$0R126$	$1L35$
17	$0L18$	$1L16$	114	$0R127$	$1R128$
18	$0L19$	$1L16$	115	$0R14$	$1R115$
19	$0L20$	$1L16$	116	$0R88$	$0L14$
20	$0L21$	$1L16$	117	$0R90$	$1L95$
21	$0L22$	$1L16$	118	$0R129$	$1R129$
22	$0L23$	$1L16$	119	$0L63$	$0R103$
23	$0R23$	$1R24$	120	$0R68$	$0R91$
24	$0R25$	$1R26$	121	$0R58$	$0R91$
25	$0R27$	$1R28$	122	$0L86$	$1R130$
26	$0R29$	$1R30$	123	$0R131$	$1R132$
27	$0R31$	$1RH$	124	$0R133$	$1R134$
28	$0L32$	$1R33$	125	$0R135$	$1R136$
29	$0R34$	$1R35$	126	$0R58$	$0R111$
30	$0L36$	$1L37$	127	$0R99$	$1R116$
31	$0R38$	$1R39$	128	$0R79$	$1R78$
32	$0R40$	$1R41$	129	$0R88$	$1R47$
33	$0R42$	$1R43$	130	$0R137$	$1R138$
34	$0R44$	$1R45$	131	$0R32$	$1R36$
35	$0R46$	$1R33$	132	$0R31$	$1R98$
36	$0R47$	$1R48$	133	$0R97$	$1R139$
37	$0R49$	$1R50$	134	$0L136$	$1R88$
38	$0R51$	$1R42$	135	$0R41$	$1L140$
39	$0R52$	$1R53$	136	$0R141$	$1R142$
40	$0R54$	$1R55$	137	$0R136$	$1R143$
41	$0R56$	$1R57$	138	$0R144$	$1R145$
42	$0R58$	$0R59$	139	$0R107$	$1R85$
43	$0R60$	$0L14$	140	$0R146$	$1R142$
44	$0R61$	$1R62$	141	$0R97$	$1R35$
45	$0R63$	$1R52$	142	$0R147$	$1R47$
46	$0R64$	$1R65$	143	$0R148$	$1R149$
47	$0L46$	$1L66$	144	$0R146$	$1R107$

State	Read 0	Read 1	State	Read 0	Read 1
48	$0R67$	$0L14$	145	$0R150$	$0L10$
49	$0R68$	$0R69$	146	$0R134$	$1R41$
50	$0R70$	$0L48$	147	$0L151$	$1L152$
51	$0R64$	$1R53$	148	$0R153$	$1R154$
52	$0L71$	$0L15$	149	$0R152$	$1R155$
53	$0R72$	$0L10$	150	$0R156$	$1R157$
54	$0R73$	$1R74$	151	$0R158$	$1R159$
55	$0R75$	$1R76$	152	$0R135$	$1R160$
56	$0R77$	$1R78$	153	$0R28$	$1R161$
57	$0R79$	$1R80$	154	$0R142$	$1R162$
58	$0L81$	$0R1$	155	$0R163$	$1R164$
59	$0R82$	$1R59$	156	$0R31$	$1R52$
60	$0L73$	$1L46$	157	$0R165$	$1R156$
61	$0R83$	$1R84$	158	$0R76$	$1R166$
62	$0R54$	$1R85$	159	$0R133$	$1R166$
63	$0L86$	$1R87$	160	$0R167$	$1R168$
64	$0R58$	$0R82$	161	$0R169$	$1R170$
65	$0R88$	$1R30$	162	$0L170$	$1R171$
66	$0R88$	$1R89$	163	$0R172$	$1R172$
67	$0L90$	$1L33$	164	$0R166$	$1R173$
68	$0L81$	$0R2$	165	$0R92$	$1R113$
69	$0R91$	$1R69$	166	$0L167$	$1L174$
70	$0R72$	$1R92$	167	$0R175$	$1R176$
71	$0R46$	$1R93$	168	$0R177$	$1R176$
72	$0R42$	$0L13$	169	$0R178$	$1R179$
73	$0R94$	$1R37$	170	$0R180$	$1R159$
74	$0R30$	$1R95$	171	$0R142$	$1R136$
75	$0R42$	$1R96$	172	$0L110$	$1R124$
76	$0R97$	$1R98$	173	$0R140$	$1R174$
77	$0R99$	$1L100$	174	$0R181$	$1R177$
78	$0R30$	$1R77$	175	$0R133$	$1R147$
79	$0R93$	$1R101$	176	$0R141$	$1R134$
80	$0R30$	$1R102$	177	$0R182$	$1R41$
81	$1R81$	$1L15$	178	$0R141$	$1R183$
82	$0R103$	$1R82$	179	$0R162$	$1R168$
83	$0R94$	$0L7$	180	$0R184$	$1R185$
84	$0R104$	$1R105$	181	$0R172$	$1R186$
85	$0R42$	$1R106$	182	$0R133$	$1R135$
86	$0R107$	$1R48$	183	$0R187$	$1R166$
87	$0R108$	$0L72$	184	$0R141$	$1R182$
88	$0R68$	$0R109$	185	$0R188$	$1R188$
89	$0R110$	$1R88$	186	$0R189$	$1R41$
90	$0R49$	$1R100$	187	$0R147$	$1R190$
91	$0R111$	$1R91$	188	$0R66$	$1L191$
92	$0R30$	$1R48$	189	$0R41$	$1R146$
93	$0R47$	$1R60$	190	$0R32$	$1R191$
94	$0L63$	$0R112$	191	$0R151$	$1R192$
95	$0R94$	$1L113$	192	$0R125$	$1R193$
96	$0R101$	$0L14$	193	$0R140$	$1R146$

B The 194-state frontier

P means a supplied ordinary PA-equivalence manuscript supports the class; A means auxiliary/search; R means explicitly rejected. This records supplied status, not a new independent proof of every older machine. Full hashes and representative paths are in `comparison/frontier194.csv`.

Name	Status	Class prefix	Construction distinction
CS both-zero6	A	`76d35bfce3a6`	Known-zero X decrement at both source26 and final emitter exit; research composition.
CSZ-A3	A	`ea9f5da94795`	CSZ plus comparator-return register3 and final-emitter register3 decrements.
CSZ-A5	A	`bb730c3c718a`	CSZ plus comparator-return register3 and final-emitter register5 decrements.
CSZ-A6	A	`3b8e1d7018aa`	CSZ plus comparator-return register3 and final-emitter register6 decrements.
Cached I-tree zero-jump seed	A	`6f9dff0d3416`	Alternate decoder plus cached controller and entry zero jump; search basis.
Gamma(D)-CSZA	A	`64c88ee1c122`	Supplementary Gamma(D) emitter plus known-zero changes.
M4R	P	`a1109e46b45a`	Reassociated coefficient4 constructor decoder; original resolved source graph.
M5	P	`debfb8a49978`	Formula constructor coefficient5; explicit variable and proof recoding.
M6	P	`41d603308985`	Formula constructor coefficient6; explicit variable and proof recoding.
MPH	A	`5e4f51fc54c4`	MP result cached in H while retaining original Shift; physical search input.
PA194	P	`47e4859f85d4`	Separate diagonal head, 67-reader decoder with seven folds, original controller.
PA194-68/8	P	`fcde56a35bab`	Same separate-head source, reassociated 68-reader decoder with eight folds.
PA194-AD	P	`387b30b4e321`	PA194 with physical A/D register exchange.
PA194A	P	`37d169082cbe`	CS entry jump at source26 becomes decrement of known-zero register A.
PA194C	P	`9d787d29cb1c`	MP result remains cached in H; Shift tests H=1; original ten-entry logical interface recompiled.
PA194CS	P	`90bca0127515`	Symmetric Shift guard; cached MP controller; contextual controller fold.
PA194CSZ = PA194X	P	`b600465cb6d9`	CS entry jump at source26 becomes decrement X; two headline names denote identical dense bytes.
PA194CZ	A	`080a545a20ef`	Cached controller; entry known-zero decrement of register 6.
PA194D	P	`c4ec894bec54`	Separate diagonal axiom uses Gamma(D), new proof selector, 69 readers and nine folds.
PA194F	P	`bcdc219de3c4`	CS final emitter jump becomes decrement of known-zero register X.
PA194H	P	`bc29bfd31443`	No-push MP, cached-only Shift; original comparison orientation.3653 source sites,4754 outcomes,57922-step startup.
PA194H6	P	`760fcde40185`	Coefficient-six constructors plus no-push MP and cache-only Shift.
PA194HA	P	`817cddc52734`	MPH cached-result MP with original sum Shift and A/D physical register exchange.
PA194HAZ	P	`aacffa8d7d9c`	HA plus known-zero final AX decrement.

Name	Status	Class prefix	Construction distinction
PA194HS = PA194NS(cache)	P	`de41942ff26f`	No-push MP, cached-only symmetric Shift;3650 source sites,4750 outcomes,57768-step startup. Two submissions deliver identical dense bytes.
PA194J	P	`aa6c94af87a9`	Ten source jumps become known-zero decrements; three physical rows change.
PA194N	P	`91e2e377b55f`	Successful MP drops dummy-zero push, retains original Shift; revised command proof compiler.
PA194N5	P	`be8cbd683d7c`	No-push MP with radix5 formula constructors;3715 source sites,4816 outcomes,58320-step startup.
PA194N6	P	`dba449969fa4`	No-push MP with radix6 formula constructors;3776 source sites,4877 outcomes,58320-step startup. Two submissions deliver identical dense bytes.
PA194NS(sum)	P	`adb95b6d6d74`	Original sum Shift guard with reversed comparison operands; no-push successful MP. This names a different machine from PA194NS(cache)=HS.
PA194Q	P	`ccf29d739b36`	Quantifier-distribution curried premises exchanged; K/S proof translation.
PA194QN	P	`f5295d775361`	Q premise-exchanged quantifier-distribution axiom plus no-push MP with original sum Shift.
PA194R	P	`aea47aaf7ffa`	Reassociated decoder of PA194F; 1111 source paths changed, same resolved source graph.
PA194SUMS	P	`6d5c5f5cceb7`	Original sum Shift guard with reversed comparison operands; original MP protocol.
PA194ZR	P	`18e996382b72`	Eight comparator returns and final emitter return become known-zero register3 decrements.
PA194altC	A	`cf97d43185a6`	Cached controller on alternate 68-reader decoder.
Rejected CS node1-X	R	`dc936f437ad4`	Native103/read1 becomes dec6; exact blank run halts at1361.
Rejected I-tree node1-X	R	`aef95573862d`	Known-zero source rewrite fails blank bootstrap; exact halt at1361.
Rejected ZR node1-X	R	`2b53937097b9`	Register6 comparator-return variant; exact blank run halts at1361.
Rejected seven-zero scanner	R	`d2f0248d6556`	Decoder local folds pass, but source-address scanner violates runtime contract.

C All 55 research deliveries

Every row records a pinned packet README reviewed for its result and scope. Full paths, exact README hashes, and the scope qualification are retained in `comparison/packet_ledger.csv`. The four current incoming archives follow the 51 placed packets.

No.	Packet	Reported contribution or limitation
1	`pa_unary_power_cover`	Unary power covering at PA strength (report of 27 October 2026). The U208 packets carry it as a baseline.
2	`U208_ConPA_Conditional_Context_2026-10-29`	U208 by a conditional tape-context reduction. States a 208-state candidate machine/u208.tm.
3	`U208_ConPA_Entry_Context_2026-10-29`	U208 by eliminating a state using entry context. States a 208-state machine machine/u208.tm.
4	`U208_ConPA_History_2026-10-29`	U208 by a finite-history decoder fold. States a 208-state candidate machine/u208.tm.
5	`U208_ConPA_2026-10-29`	U208 by a three-cell fold. States a 208-state candidate machine/u208.tm.
6	`u208_research_2026-10-29`	U208 research release. States two 208-state machines, machine/u208.tm and machine/variant_plain/u208.tm.
7	`Below_U208_Research_2026-10-29`	Research toward fewer than 208 states. States that no machine with 207 states or fewer was obtained.
8	`u208_comparison_research_2026-10-29`	Comparison of the 208-state proposals and a search below 208. States that the target was not achieved.
9	`ConPA_below208_research_2026-10-29`	Comparative audit and reduction search below 208. States that it does not establish a 207-state machine.
10	`pa_consistency_research_2026-10-29`	Audit and new source reductions. States that no validated machine below 208 states was obtained.
11	`pa208_research_2026-10-29`	Comparison and research below 208 states. States that it does not establish a valid machine below 208.
12	`pa_tm_research_2026-11-13`	Comparison and reduction research. States a new 196-state variant with shorter source paths and nothing below 196.
13	`pa_consistency_research_2026-11-13`	Comparison and diagonal-vacuity research. States a shorter axiom program with a 199-state implementation and nothing below 196.
14	`conpa_tm_comparison_and_reduction_search`	Comparison and reduction search. States two experimental 196-state command variants and nothing below 196.
15	`pa196_research_2026-11-13`	PA196 comparison, source factorization and proof obligations. States a source tree of 157 internal nodes and nothing below 196.
16	`pa_consistency_research`	Research and verification packet. States that no machine below 196 states has been validated.
17	`pa_tm_comparison_and_reductions`	Comparison and new rewrites, a research snapshot.
18	`pa_tm_extended_research_2026-11-13`	Comparison and refined reduction search. States that the smallest accepted count is 196.
19	`conpa_tm_comparison_and_search`	Comparison and a search below 196. States that none was obtained; files named 195 are rejected or diagnostic candidates.
20	`pa_consistency_reduction`	Comparison and reduction experiments. States two new 196-state variants and a shorter source program with a 203-state implementation.
21	`pa_consistency_comparison_2026_11_13`	Comparison and new reduction experiments. States that nothing below 196 was obtained.
22	`PA_consistency_TM_research_2026-11-13`	Comparison and new source reductions. States that no machine below 196 is claimed.

No.	Packet	Reported contribution or limitation
23	pa_consistency_research_2026-11-13_2	Comparison and search below 196. States two new layouts with 196 and 198 states and a command-level reduction.
24	conpa_tm_study_2026-11-13	Comparison and reduction study. States that nothing below 196 was obtained.
25	pa_below196_research	Comparison and reduction experiments. States that nothing below 196 was obtained.
26	pa_consistency_comparison_and_new196_2026-11-13	Comparison and reduction research. States that it contains no accepted machine below 196.
27	pa_tm_research_2026-11-13_2	Comparison and reduction investigation. States same-size work, failed candidates and nothing below 196.
28	pa_consistency_cross_archive	Cross-archive constructions and reduction search. States that it does not certify a 195-state machine.
29	pa195_conpa_2026-11-13	PA195. States a 195-state machine pa195/pa195.tm and an article with a PA proof of the equivalence.
30	pa_consistency_reduction_audit_2026-11-14	Comparison and reduction audit. States nothing below 196 and no new theorem.
31	pa_consistency_research_2026-11-14	Comparison and NEG18. States an 18-instruction encoded negation with a compiled machine of 197 states.
32	PA195_ConPA_research_2026-11-14	A 195-state machine. States the primary machine_195S.tm and a second table machine_195C.tm.
33	pa195_followup_research	Comparison and searches below 195. States a graph-distinct 195-state machine M195S8 and nothing below 195.
34	pa194_conpa_2026-11-13	A 194-state machine. States the table construction/tables/pa194.tm and a proof of the equivalence in PA.
35	pa194_conpa_2026-11-13_2	PA194. States a 194-state machine pa194/pa194.tm and an article with a PA proof of the equivalence.
36	pa195_diagonal_combined_2026-11-13	PA195DC. States a different 195-state machine.
37	pa194C_conpa_2026-11-13_2	PA194C. States a materially different 194-state machine, an article with a PA proof of the equivalence, and nothing below 194.
38	pa194_new_designs_2026-11-13	Two new 194-state machines, PA194C and PA194CS. States that nothing below 194 was obtained.
39	pa194c_conpa	PA194C, a cached-head 194-state machine. States that no machine below 194 was established.
40	pa194c_conpa_2026-11-13_1	PA194C. States a different 194-state machine and an article with a PA proof of the equivalence.
41	pa194c_conpa_2026-11-13	PA194C, a cached-head 194-state machine. States an article with a PA proof of the equivalence and nothing below 194.
42	pa194_radix_family_2026-11-13	New 194-state machines calibrated to Con(PA), a radix family.
43	pa194j_pa194d_conpa_2026-11-13	PA194J and PA194D. States two new 194-state machines.
44	pa194q_conpa_2026-11-13	PA194Q and PA194CSZ. States two new 194-state machines and that no machine below 194 was established.
45	PA194R_research_2026-11-14	PA194R. States a different 194-state machine with a new decoder layout and that no machine below 194 was verified.
46	pa194N_conpa_2026-11-13	PA194N. States a new 194-state machine, an article with a PA proof of the equivalence, and nothing below 194.
47	pa194zr_conpa_2026-11-13	PA194ZR. States a new 194-state machine machine/pa194zr.tm, a PA proof of the equivalence, and nothing below 194.

No.	Packet	Reported contribution or limitation
48	`pa194hs_conpa_2026-11-13`	PA194H and PA194HS: no-push successful MP returns in top; Shift tests cache only. HS uses symmetric comparison; HS is byte-identical to NS from a separate submission. Both194 states, ordinary PA proof and two physical symbolic checkers reported.
49	`pa194ns_conpa_2026-11-13`	PA194NS is byte-identical to PA194HS; independent submission gives full command compiler and universal implementation checks.194 states; no sub194 result.
50	`pa194n5_n6_conpa_2026-11-13`	PA194N5 and PA194N6 combine radix5/6 formula constructors with no-push successful MP. Each194 states; standard-library JavaScript universal verifier and ordinary PA proof. N6 duplicates separate N6 research delivery. PDF was typeset directly, not compiled from TeX.
51	`pa194n6_research_2026-11-13`	PA194N6 duplicates the N6 machine in the N5/N6 packet, with fuller independent physical replay, source-splice review and command/radix proof;194 states, no sub194 result.
52	`PA194NS_research`	NS(sum) and SUMS preserve the original t+h=1 Shift guard but reverse comparison operands; NS(sum) also drops successful-MP dummy push. Distinct from the earlier cache-only NS=HS naming collision. Reports two 194-state constructions, inherited numerical-calculus calibration and Python/JavaScript symbolic audits; no sub194.
53	`pa194h6_conpa_2026-11-13`	H6 combines coefficient-six formula encoding, no-push MP and cache-only Shift. Reports ordinary PA proof, complete source binding, two universal symbolic physical checkers and startup; 194 states, no sub194.
54	`pa194haz_research_2026-11-14`	HA and HAZ combine MPH cached-result MP with A/D physical register exchange; HAZ additionally replaces final AX jump by known-zero decrement. Reports inherited calibration and arithmetic-box-based Python/JavaScript checks. Two 194-state tables, no sub194.
55	`pa194qn_research_2026-11-13`	QN combines Q premise-exchanged quantifier-distribution axiom with no-push MP and original sum Shift. Reports ordinary PA proof and two universal physical checkers. 194 states, no sub194; two 195-state MP-tail experiments lack completed new consistency proofs.